PVA

Contents

Research Summary....2

Introduction....3

- Context Analysis....3
- Stakeholder Insights....5
- Literature Overview....5
- Identified Gap....7
- Research Question....8
- Scope....8

Background....8

- Relevant literature on techniques....8
- related work comparison....8
- AI suitability justification....8
- ethical and societal considerations....8

Methodology Research Design....8

- Planned research design....8
- Type of study....8
- Baseline....8
- Comparison logic....8

Methodology Requirements....9

- Legal Requirements....9
- Ethical Requirements....9
- Organizational Requirements....9
- Functional Requirements....9
- Technical Requirements....9

Methodology Data....9

- Data Sources....9

Data characteristics....9
Data risks....9
Bias consideration....9
Methodology Model & Architecture....9
Proposed techniques....9
alternatives considered....9
literature-based justification....9
expected trade-offs....9
Methodology Evaluation....10
Defined metrics....10
Justification....10
Validation strategy....10
Baseline....10
Definition....10
stakeholder evaluation plan....10
Results....10
Discussion....10
Anticipated limitations....10
risk analysis....10
planned validity and reliability safeguards....10
Conclusion....10
Clear objective....10
Deliverables....10
Realistic planning and milestones....10
Data....10
Planning....11
Bronnen....11

Research Summary

... <soort van abstract hier>

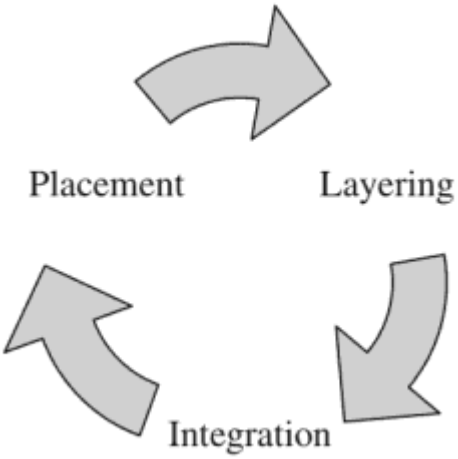
Introduction

Context Analysis

Money Laundering

Money laundering is the process of concealing the illegal origin of funds to make them appear legitimate, allowing criminals to use the proceeds in the legal economy [1], [2]. These funds often come from crimes such as drug trafficking, fraud, or illegal pornography, tax evasion, robbery, blackmail, bribery, piracy and people trafficking [1], [3]. Once obtained, the money cannot be spent freely, so criminals must disguise its illegal origin. This is achieved through the money-laundering process, which typically occurs in three stages: placement, layering and integration [3].

<this citation is the same source but from different pages. Should i add another citation? Or keep it one citation and update the pages inside the citation?>



Placement

Placement is the first step in the laundering process. In this step, criminals attempt to introduce their funds into the financial system. This is often associated with depositing cash into bank accounts, but placement is not

limited to plain deposits. Some examples of introducing criminal money to the financial system cover: purchasing artwork, antiques and lottery tickets.

Layering

Once criminally earned funds have been introduced to the financial system, criminals try to hide the origin of this money. This can be done in various ways, such as investing in legitimate assets, buying and selling collectibles, or making multiple smaller transactions across different accounts. The more these funds are moved back and forth, the harder it becomes to trace their original source.

Integration

Once the origin of illegal funds is difficult to trace, criminals can freely integrate their laundered money into the economy. This can be done by investing in businesses, purchasing property, buying jewellery, or other assets that appear legitimate.

Money Laundering Detection

Money laundering detection or AML (anti-money laundering) refers to the process of detecting money laundering. AML technology has changed significantly over the past years. Evolving from dedicated AML teams to semi-automized machine learning workflows [4].

Limitations of Traditional AML Methods

Initially, companies relied on dedicated AML (Anti-Money Laundering) departments where employees were trained to manually identify suspicious transactions. With the growth of transaction volumes, rule-based systems became one of the first forms of automated detection. These systems typically monitored transactions and generated alerts for activity exceeding thresholds, such as amounts over $10,000, amounts just below $10,000, or multiple smaller transactions within 24 hours that collectively surpassed $10,000. While rule-based systems improved efficiency, they had limitations in handling complex laundering patterns and large-scale data, which led to the adoption of more advanced methods, including graph analytics, machine learning, and deep learning techniques.

Machine Learning in AML

Machine learning is often proposed as an alternative to traditional rule-based AML systems because of its ability to model complex patterns. Rule-based approaches rely on manually defined rules and fixed thresholds, which makes them less flexible and more likely to produce many false positives and struggle when patterns change over time. In contrast, machine learning models can learn complex relationships from data and adapt to new laundering strategies over time. This dynamic nature makes machine learning particularly suitable for detecting sophisticated and previously unseen laundering techniques that would not be captured by predefined rules.

Prior literature shows that a wide range of machine learning approaches have been applied to AML. Most studies rely on supervised learning [5]. However, due to the scarcity and noisiness of labelled AML data, unsupervised and semi-supervised methods have also received attention.

Reinforcement Learning in AML

Little research has been conducted on reinforcement learning in AML. Typically, reinforcement learning requires an environment that an agent can interact with, which is not naturally present in AML settings. Instead, AML research mostly relies on synthetic datasets rather than interactive environments. However, with creative engineering, it is possible to simulate an environment that allows an agent to interact with the data, making reinforcement learning a potentially interesting direction for further research.

Abstract. Money laundering (ML) is the process by which criminals convert illicit funds gained from drug sales, human trafficking, and other illegal operations into legal funds that can later be spent freely and without restrictions. The annual amounts of money involved in ML are in the billions of dollars, and it is believed that it accounts for 2% to 5% of worldwide GDP [10] ~ copy pasted from **The 3rd International Conference on Artificial Intelligence and Computer Vision (AICV2023), March 5–7, 2023** Pagina 95. Consider of dit waarde toevoegt. Zoja herschrijf in je eigen woorden

1. FIU-the Netherlands. (2023). What is money laundering? Retrieved from FIU-Nederland: https://www.fiu-nederland.nl/en/home/what-is-money-laundering/

2. Financial Crimes Enforcement Network. (n.d.). What is money laundering? Retrieved from FinCEN.gov: https://www.fincen.gov/what-money-laundering

3. Cox, D. (2012). Handbook of Anti-Money Laundering. In D. Cox, Handbook of Anti-Money Laundering (pp. 5-13). John Wiley & Sons.

4. Weber, M. (2018). Scalable Graph Learning for Anti-Money Laundering: A First Look.

5. Nazanin Bakhshinejad, R. S. (2022). A Survey of Machine Learning Based Anti-Money Laundering. (not peer reviewed, but its a survey so is that ok? They don't really claim anything they just share what has been done)

Stakeholder Insights

Wie zijn de stakeholders?

- **Medewerkers** die moeten bellen bij fraudeleuse alarmen.
- **Bedrijven** die in de knoei kunnen komen met model bias en explainability
- **Klanten** die een transactie onterecht niet kunnen afronden omdat de transactie fraudeleus lijkt.
- **Klanten** die terecht transacties terecht niet kunnen afronden omdat zij geld proberen te witwassen.

Maak ook zo een stakeholder diagram

Vragen voor data scientist ING?

- Kun je iets vertellen over de evolutie van de modellen die jullie hebben gebruikt voor witwasdetectie over de jaren?
  Wat voor system/technologie/algoritme gebruiken jullie nu?
- Kun je iets kwijt over hoe goed het model performt? (als het niet vanzelf naar boven komt: En welke metriek gebruiken jullie om het model te evalueren?)
- Welke technieken hebben jullie allemaal gebruikt?
- Wat zijn de grootste problemen omtrent AML binnen ING
- Kun je iets vertellen over hoe de TMNL samenwerking impact heeft Gehad op jullie werkwijze? (Documenteer iets over tmnl)
- Limiteert de explainability regels jullie model keuze?

- Hoe zorgen jullie ervoor dat jullie je aan de AVG wet houden

<Interview a stakeholder. Probably someone from DLL>

Literature Overview

This section provides an overview of the current state-of-the-art regarding Machine Learning applications within Anti-Money Laundering.

The IEEE-CIS Fraud Detection Competition

In 2019, Kaggle hosted the *IEEE-CIS Fraud Detection Competition* in collaboration with Vesta Corporation, an event that remains a foundational milestone for machine learning in the financial sector. By providing a massive, real-world dataset directly from Vesta's e-commerce transactions, the competition set a definitive benchmark for the state-of-the-art in transaction monitoring and fraud detection algorithms.

Vesta, a leader in guaranteed e-commerce payment solutions, shared this data to challenge researchers to improve the effectiveness of fraudulent transaction alerts. The competition consisted of two main goals: to increase the detection of criminal activity while minimizing the "false positives" that lead to legitimate customers having their cards declined at the checkout.

The competition allowed researchers to join individually or in teams, with the flexibility to submit an unlimited number of solutions to refine their results. The scale of the competition is reflected in the final statistics:

- **Participants:** 7,389 individual researchers.
- **Collaboration:** 6,351 unique teams.
- **Intensity:** A total of 125,219 submissions were analyzed.
- **Prize:** A $20,000 prize pool was awarded to the top-performing solutions.

The competition results provide a clear map of what defines as the "state-of-the-art" for tabular fraud data. An analysis of the top-performing teams (which shared their solution) reveals a heavy reliance on supervised learning, specifically Gradient Boosting Decision Trees (GBDT):

- **1st Place (Score: 0.9677):** The winning solution utilized a stacked ensemble of CatBoost, XGBoost, and LightGBM. The team also developed a Neural Network, but it was dropped from the final submission because

it underperformed compared to the Gradient Boosting models. [1st-place-model-src].

- **2nd Place (Score: 0.9672):** This team achieved their ranking using a blend of LightGBM, CatBoost, and XGBoost. Their performance was further boosted by a temporal post-processing step that overrode predictions based on a user's past fraud history [2nd-place].
- **5th Place (Score: 0.9424):** The 5th place solution relied on a simple average of four independent LightGBM models. They also created a specialized model focused on "single transaction" users, the output of which was used as a feature for their main LightGBM ensemble [5th-place].

[https://www.kaggle.com/competitions/ieee-fraud-detection/overview]

State of the art kan ver uitgebreid worden als het nodig is met deze bron: **The 3rd International Conference on Artificial Intelligence and Computer Vision (AICV2023), March 5–7, 2023** pagina 95+. (zou state of the art updaten van 2019 naar 2023)

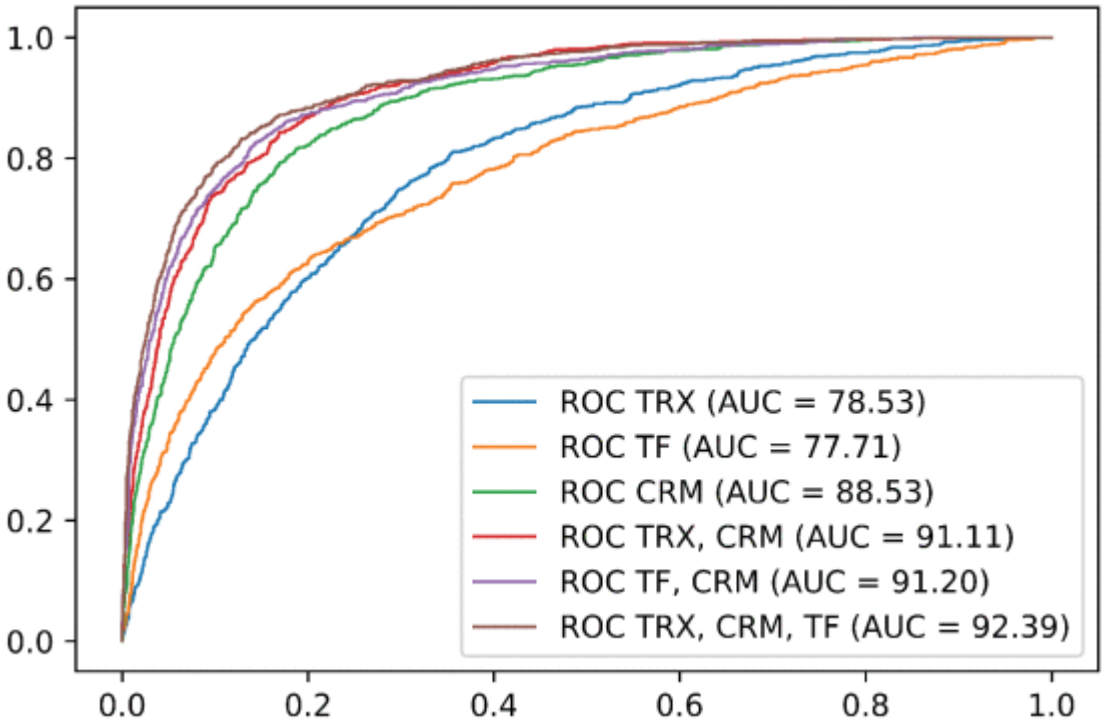
State-of-the-Art

[Investigation of Applying Machine Learning for Watch-List Filtering in Anti-Money Laundering | IEEE Journals & Magazine | IEEE Xplore] compares SVM, NB and DT algorithms. SVM outperforms others.

Algorithm	Accuracy Avg	Precision Avg	Recall Avg	Accruacy Standard Deviation
SVM_6	0.848	0.851	0.848	0.023
NB_4	0.824	0.827	0.824	0.019
DT_4	0.783	0.786	0.783	0.027

A Time-Frequency Based Suspicious Activity Detection for Anti-Money Laundering | IEEE Journals & Magazine | IEEE Xplore Random Forest with different features to compare which ones are more important

Training Set	FPR	FNR	PPV	TPR	F-Score
Transaction Features	28.10%	25.30%	49.26%	74.70%	59.37%
Time Frequency Features	14.90%	41.02%	59.11%	58.98%	59.05%
CRM Features	23.52%	20.93%	55.11%	79.07%	64.95%
Transaction and CRM Features	18.76%	19.75%	60.97%	80.25%	69.29%
Time Frequency and CRM Features	11.42%	25.85%	70.33%	74.15%	72.19%
Transaction, CRM and Time Frequency Features	11.85%	22.10%	70.59%	77.90%	74.06%



Research papers using real banking data Model Choice

Methods used most (2017 - 2023)

SVM [3], [7], [14],

Random Forest [4], [7], [8], [14], [23],

XGBoost [6], [23], [15], [19]

Logistic Regression [7], [8], [14], [23], [15], [19]

Methods explored less (2017 - 2023)

ANN

GraphDL [8]

GraphCNN [21]

Reinforcement Learning [-] – tot aan 2023 is er geen RL research gedaan (op echte datasets) dit kan een reden zijn om RL verder te gaan onderzoeken (dus gap)

Reinforcement Learning State-of-the-Art

What are the best performing models and approaches in reinforcement learning within AML

Hoe goed zijn RL modellen vandaagderdag. Tot aan 2026

...

Identified Gap

Wat is een interessante Gap?

- ➔ Erg veel technieken zijn nog niet veel explored
- ➔ RL is ook weinig explored

Kan RL een gap zijn?

The gap of RL literature in AML

Research Question

Research questions?

Scope

What will be in scope?

What will be outside scope?

Background

Relevant literature on techniques

related work comparison

AI suitability justification

ethical and societal considerations

Methodology Research Design

Planned research design

Type of study

Baseline

Comparison logic

Methodology Requirements

Legal Requirements

Ethical Requirements

Organizational Requirements

Functional Requirements

Technical Requirements

Requirements must be derived from field + desk research

Methodology Data

Data Sources

<er zijn volgens research papers wel datasets online beschikbaar die niet syntetisch zijn dus dit moet allemaal waarschijnlijk opnieuw>

For my dataset I have considered multiple widely known AML datasets, such as the IBM dataset [1], SynthAML[2] and SAML-D [3]. However the issue with each dataset is that labels are absent. For reinforcement learning, rewards are a neccessity for the model to understand when it is performing well. If there are no labels indicating which transaction are laundred, it will be very difficult to determine when the model should be rewarded or not. Since rewards are a core concept within reinforcement learning, it won't be possible to train an agent without well designed rewards.

Mention that all datasets in aml are synthetic -> klopt dus wss ni

There is one dataset that contains the column is_fraud. However, this dataset consists of only 2430 rows [4]. Instead of proceeding with a small dataset, I decided to create my own dataset using tools like: AMLSim [5] or AMLGentex [6]. These tools are designed to created syntehtic datasets for AML.

1. IBM Transactions for Anti Money Laundering (AML)
- https://www.kaggle.com/datasets/ealtman2019/ibm-transactions-for-anti-money-laundering-aml

2. SynthAML Dataset
- https://springernature.figshare.com/collections/SynthAML_a_Synthetic_Data_Set_to_Benchmark_Anti-Money_Laundering_Methods/6504421/1

3. SAML-D Dataset
- https://www.kaggle.com/datasets/berkanoztas/synthetic-transaction-monitoring-dataset-aml

4. Money Laundering Data
- https://www.kaggle.com/datasets/maryam1212/money-laundering-data/data

5. AMLSim - https://knowledge.iadb.org/en/knowledge-resources/code-development/open-source-solution/amlsim?utm_source=chatgpt.com

6. AMLGentex - https://github.com/aidotse/AMLGentex

Data characteristics

Data risks

Bias consideration

Methodology Model & Architecture

Proposed techniques

alternatives considered

literature-based justification

expected trade-offs

Methodology Evaluation

Defined metrics

Justification

Validation strategy

Baseline

Definition

stakeholder evaluation plan

Results

Definition of what will count as success; measurable criteria and thresholds

Discussion

Anticipated limitations

risk analysis

planned validity and reliability safeguards

Conclusion

Clear objective

Deliverables

Realistic planning and milestones

Planning

Bronnen

Vragen voor Michiel

Moet ik de state of the art uitbreiden? Tot nu toe tot 2019